

Daniel Hardesty Lewis

AI Consultant – ML systems, HPC, & scientific computing

AI consultant specializing in ML systems engineering, HPC infrastructure, and bridging research to production. Hands-on experience shipping deep generative models, orchestrating distributed workloads at supercomputer scale, and teaching ML, CUDA, and C++ at the graduate level.

Core Expertise

- 8 years **ML Systems Engineering**: Build and operate ML training and inference pipelines, including deep generative models, rigorous evaluation, and automated retraining
- 8 years **HPC & Distributed Computing**: Scale batch workloads across leadership-class clusters, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 10 years **Scientific Computing**: Spatiotemporal modeling, numerical methods, and simulation-based decision-support across geospatial, financial, and climate domains
- 8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers and researchers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infrastructure	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
Systems	CUDA; C++; MPI; OpenMP; SLURM; containerized HPC

Experience

Professional

- Nov. 2024 – Present **Co-founder**, [Homecastr](#), New York, NY
 - Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.
 - Specified an **agentic workflow** for BOM parsing and tariff simulation.
 - Directed a **conversational AI interface** enabling non-technical decision-makers to retrieve BOMs and receive proactive compliance alerts without engineering support.
 - Ran **customer discovery and live demos** with Deloitte and ScoutyAI; **scoped integration requirements** before presenting the prototype to KPMG.
- Nov. 2024 – Present **Co-founder**, [PoliBOM](#), Seattle, WA
 - AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation
 - Specified an **agentic workflow** for BOM parsing and tariff simulation.
 - Directed a **conversational AI interface** enabling non-technical decision-makers to retrieve BOMs and receive proactive compliance alerts without engineering support.
 - Ran **customer discovery and live demos** with Deloitte and ScoutyAI; **scoped integration requirements** before presenting the prototype to KPMG.
- Nov. 2023 – Present **Founder**, [Summit Geospatial](#), New York, NY
 - High-resolution elevation data and web delivery for engineering and hazard workflows.
 - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
 - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

500 RIVERSIDE DR – NEW YORK, NY 10027 – UNITED STATES

☎ +1 (713) 371-7875 • ✉ daniel@homecastr.com • 🌐 dlewis.ai • in [dhardestylewis](#)
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- Aug. 2021 – **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Aug. 2023
- Co-founded [summitgeospatial.com](#) to commercialize and sustain state-level terrain models developed for the \$40M [Texas Disaster Information System \(TDIS\)](#), delivering precision decision-support for civil engineering.
 - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating [Real-Time Inundation Mapping](#) and TDIS.
 - Scaled climate and flood models on supercomputers, orchestrating 800,000-node distributed jobs across containerized HPC clusters while managing multi-million-dollar compute budgets.
 - Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
 - Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
 - Contributed fundamental research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#), one of geomorphology's highest honors.
 - Served as Technical Reviewer for [Frontera Doctoral Fellows](#) (2023) and [Large-Scale Computing Pathways](#) (2022) applications, evaluating \$10M+ in computing resource allocations on one of the world's largest supercomputers.
- Feb. 2018 – **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Received research recognition from AAAI President Yolanda Gil during her [farewell address](#) for contributions to automated scientific modeling.

Research

- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirs** (**Industry-sponsored research**), *Columbia University*, New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- July 2019 – **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

- Jan. 2018 – **Co-instructor**, *The University of Texas at Austin*, Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TIS*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

Education

- expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY
- 2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

Key Projects

- oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#) Bikepacking US West & East Coasts

Languages

- Spanish Native Proficiency French Professional Working Proficiency

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